

<https://github.com/20210330-12/COMP3011-Coursework1>

Football Analytics & Community REST API

COMP3011 – Web Services and Web Data

Name

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Project Motivation

Problem

Football statistics are widely available but often fragmented across multiple platforms.

Goal

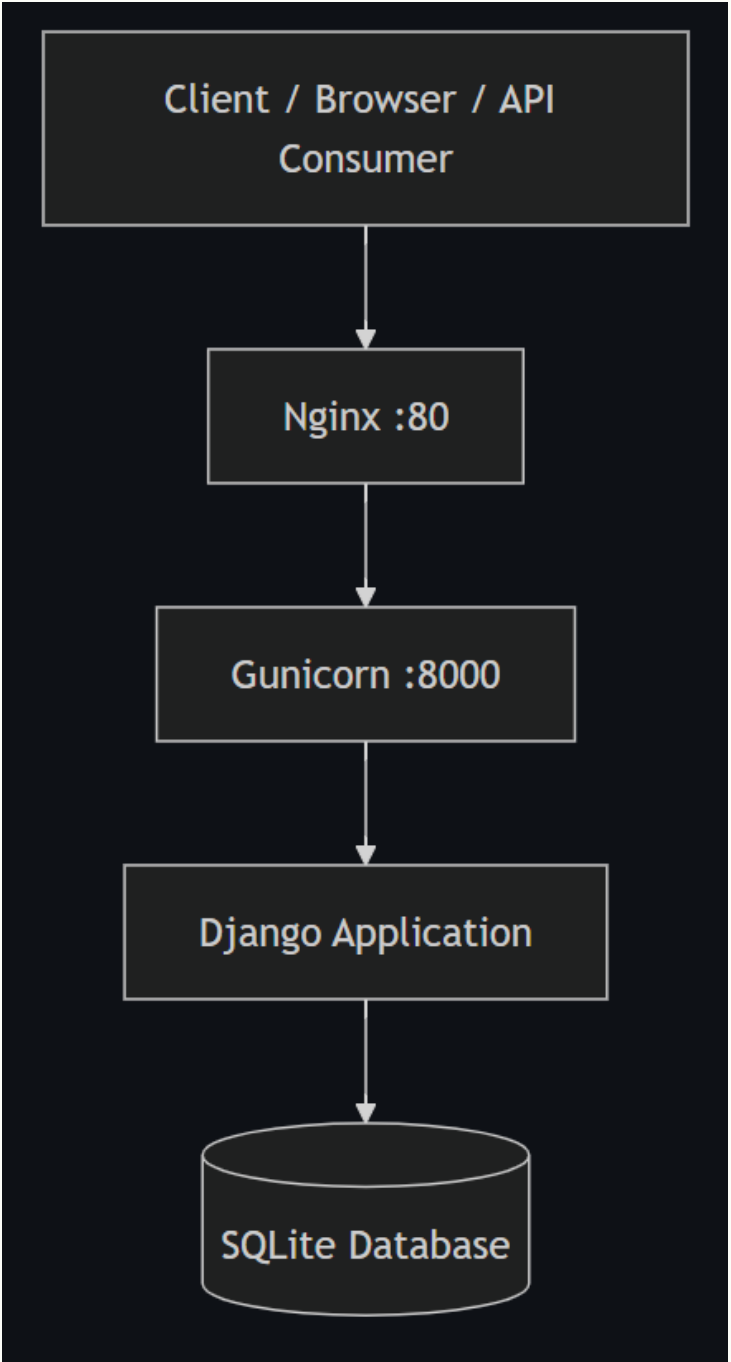
Create a unified REST API that provides:

- football analytics
- community interaction

Key Idea

Combine football analytics and community interaction in a single API platform.

System Architecture & Database Design



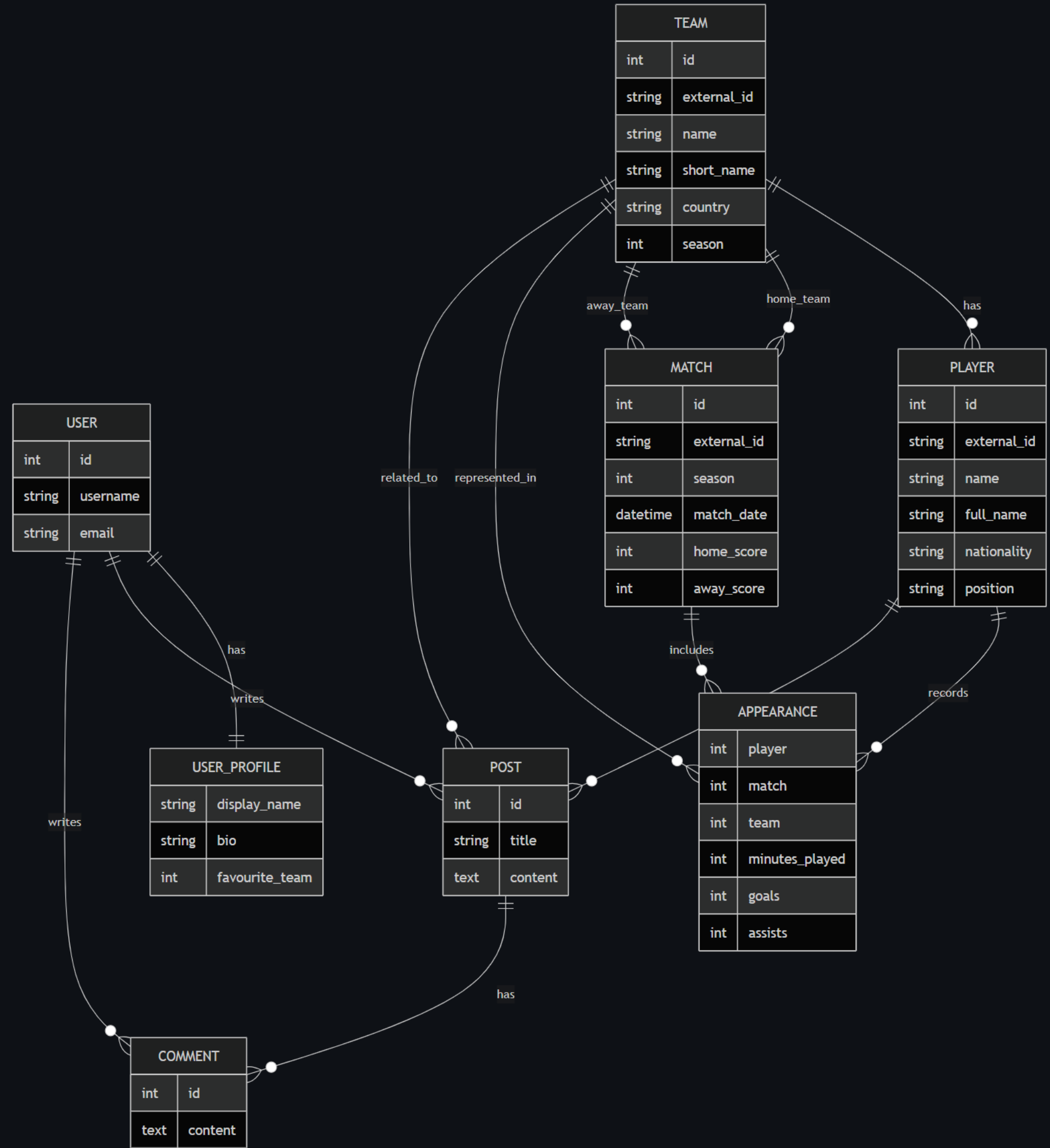
AWS EC2

Deployment challenge:
EC2 t3.nano memory limitation

Solution:
Reduced Gunicorn workers

← Production
Deployment
Diagram

ERD →



API Features & Analytics Example

Main API Features

- Football data endpoints
- Player statistics
- Match information
- Analytics endpoints
- Community posts and comments
- Token authentication

Analytics

ex. League standings calculation

Matches → appearances → aggregated statistics

- Win = 3 points
- Draw = 1 point
- Loss = 0 points

Conclusion

Key Outcomes

- RESTful football analytics API
- Community discussion system
- Real dataset integration
- External deployment

Future Work

- Advanced analytics
- Frontend interface
- Live sports data
- Performance optimization (caching)

Thank you, Questions?

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